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Integrating tracer-based metabolomics data and metabolic fluxes in a linear fashion via Elementary Carbon Modes

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ABSTRACT

Constraints-based modeling is an emergent area in Systems Biology that includes an increasing set of methods for the analysis of metabolic networks. In order to refine its predictions, the development of novel methods integrating high-throughput experimental data is currently a key challenge in the field. In this paper, we present a novel set of constraints that integrate tracer-based metabolomics data from Isotope Labeling Experiments and metabolic fluxes in a linear fashion. These constraints are based on Elementary Carbon Modes (ECMs), a recently developed concept that generalizes Elementary Flux Modes at the carbon level. To illustrate the effect of our ECMs-based constraints, a Flux Variability Analysis approach was applied to a previously published metabolic network involving the main pathways in the metabolism of glucose. The addition of our ECMs-based constraints substantially reduced the under-determination resulting from a standard application of Flux Variability Analysis, which shows a clear progress over the state of the art. In addition, our approach is adjusted to deal with combinatorial explosion of ECMs in genome-scale metabolic networks. This extension was applied to infer the maximum biosynthetic capacity of non-essential amino acids in human metabolism. Finally, as linearity is the hallmark of our approach, its importance is discussed at a methodological, computational and theoretical level and illustrated with a practical application in the field of Isotope Labeling Experiments.

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1. Introduction

Systems biology aims to create *in silico* models capable of simulating all the components of a cell working in a synergistic fashion (Kitano, 2002). To achieve this goal it is essential to know the performance of all the components at different levels of complexity (DNA–RNA–protein–metabolites), as well as their underlying interactions. This ambitious aim requires continuous communication between biological experiments and *in silico* modeling techniques, the last one being where the work presented here is focused.

Among the different levels of complexity, the one that best represents the phenotype is the metabolic level (Fiehn, 2002). Analysis of cellular metabolism has for decades allowed the diagnosis of diseases (Warburg, 1956), as well as the elucidation of novel drug targets (Boros et al., 2004; Cascante et al., 2002). The effort in reconstructing metabolic networks in different organisms (Kanehisa and Goto, 2000; Ma et al., 2007; Schellenberger et al., 2010), along with advances in "omics" disciplines (Alizadeh et al., 2000), has allowed the analysis of cellular metabolism in a more systematic fashion.

The activity of cellular metabolism is quantified by the fluxes of its inherent enzyme-catalyzed reactions. The knowledge of these fluxes is crucial to model cell behavior, however the vast majority of them cannot be directly measured through experiments. Constraint-based modeling (CBM) has emerged as a promising tool for determining and analyzing fluxes (Price et al., 2004). The foundations of CBM were laid in the early 1990s with the stoichiometric Metabolic Flux Analysis (MFA) (Varma and Palsson, 1994). In essence, stoichiometric MFA constructs a linear system of equations based on the assumption that the metabolic network is at steady-state and uses an experimental measure of some uptake/secretion fluxes of (extracellular) metabolites. Unfortunately, this linear system of equations is typically under-determined. For this reason, different tools emerged to explore the solution space of the metabolic network and its underlying properties. The most common method in CBM is Flux Balance

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Analysis (Kauffman et al., 2003), which assumes that the fluxes in the network follow a biological objective function to be optimized, *e.g.* maximizing cellular growth. In order to refine predictions made via CBM, a crucial task in Systems Biology is the development of novel methods integrating high-throughput experimental data (Chandrasekaran and Price, 2010).

Isotope Labeling is the most widely used experimental technique to infer metabolic fluxes. Isotope Labeling Experiments (ILEs) are based on the introduction of substrates labeled with isotopic tracers at specific locations. The inter-conversion of substrates via (intracellular) enzyme-catalyzed reactions presents asymmetries at the atomic level (Stephanopoulos et al., 1998). These asymmetries are found in the steady-state isotopic enrichment of metabolites, which is represented with the concept of isotopomers, *i.e.* isomers with isotopic atoms. Isotopomers may involve different isotopic tracers. We will here focus on ^{13}C, which is the most established in the analysis of metabolic networks.

The experimental measurement of the isotopomer distribution of certain metabolites using a metabolomics platform (Nuclear Magnetic Resonance or Mass Spectrometry) allows us to infer intracellular fluxes (Matsuoka and Shimizu, 2010; Sauer, 2006; Selivanov et al., 2010; Shimizu, 2002; Tang et al., 2009; Wiechert, 2001). Different theoretical frameworks have been developed to carry out this task (Antoniewicz et al., 2007; Schmidt et al., 1997; Selivanov et al., 2006, 2005, 2004; Wiechert et al., 1999; Zupke and Stephanopoulos, 1994). These approaches result in non-linear sets of equations, which require the use of non-linear optimization techniques (deterministic, meta-heuristic and/or stochastic) to compute a solution (Glover and Kochenberger, 2003). In general, such methodologies incur high computational costs and most of them do not guarantee global optimality. Global optimization techniques, which guarantee the robust computation of global optima do exist (see *e.g.* Floudas and Stein, 2007; Lin et al., 2005), but they are normally computationally costly. For this reason, these methods have been typically applied to small-sized (simplified) metabolic networks. Active research is currently devoted to integrate ILEs into larger networks, namely building novel modeling frameworks (Suthers et al., 2007, 2010) and algorithms to deal with their inherent non-linearity (Sokol et al., 2012). However, further work is still needed to achieve this objective for genome-scale metabolic networks (GSMNs), which are at the core of CBM and Systems Biology.

In this context, different attempts have been made to integrate ILEs into CBM, *e.g.* Vo and Palsson (2006). However, the non-linear nature of the constraints arising from ILEs makes this task difficult, requiring a drastic change in the structure of the problem (Wiechert, 2001). Note here that an important ingredient in the success of CBM is its underlying linearity. Indeed, most of the tools in CBM are built on linear (integer) optimization. Therefore, for a more natural integration of ILEs and CBM, linearity is essential. In this article, we show that this integration can be achieved via Elementary Carbon Modes (ECMs), a generalization of Elementary Flux Modes (EFMs) at the carbon level (Pey et al., 2011). Based on this concept, we present a novel framework that naturally extends fundamental constraints on CBM. In particular, aside from using the experimental measurement of transport fluxes of certain extracellular metabolites (in bold in Fig. 1A), as typically used in CBM, we provide formalism to integrate tracer-based metabolomic data of extracellular metabolites in a linear fashion (Fig. 1B), which further restricts the steady-state flux solution space.

2. Methods

In this section we present details concerning a novel set of linear constraints linking tracer-based metabolomics data and metabolic fluxes via Elementary Carbon Modes (ECMs). For this purpose, we first review the concept of ECMs. We then show how to integrate ECMs and tracer-based metabolomics data, as well as ECMs and metabolic fluxes. Finally, these novel constraints are discussed and integrated in the framework of CBM and GSMNs.

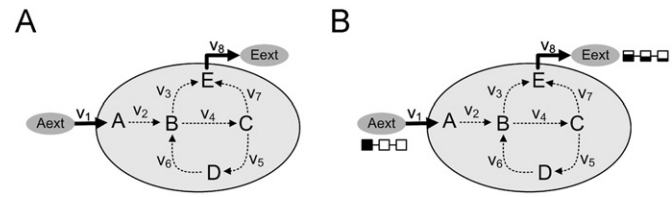

Fig. 1. (A) Scheme of experimental data used in standard CBM and (B) our approach.

2.1. Elementary Carbon Modes

Consider a metabolic network that comprises R reactions and C metabolites. Each reaction is associated with a flux variable v_r, $r=1,\ldots,R$ where $v=[v_1,\ldots,v_r]$ is the flux vector. The set of reactions is typically classified into reversible (Rev) and irreversible (Irr) reactions. Fluxes in irreversible reactions satisfy non-negativity constraints, as observed in

$$v_r \geq 0, \quad r \in Irr \tag{1}$$

The set of metabolites is divided into two subsets, namely external (E) and internal (I) metabolites. For internal metabolites it is assumed that no accumulation or depletion is possible, therefore the steady-state condition holds (see Eq. (2)). s_{cr} is the stoichiometric coefficient associated with compound c ($c=1,\ldots,C$) in reaction r ($r=1,\ldots,R$). External metabolites are typically considered as sources/sinks and (sometimes) as cofactors.

$$\sum_{r=1}^{R} s_{cr} v_r = 0, \quad c \in I \tag{2}$$

For illustration, Fig. 2A shows an example metabolic network, which involves 7 metabolites and 8 reactions. In this toy example all reactions are set irreversible and metabolites A, B, C, D and E are internal.

An elementary flux mode (EFM) is a minimal subset of reactions able to carry flux in steady state with all irreversible reactions in the appropriate direction (Schuster et al., 2000).“Minimal” means here that no subset of reactions in an EFM is able to carry flux in steady state. This simplicity condition was termed the non-decomposability condition (Schuster and Hilgetag, 1994). For the example network in Fig. 2A, 3 EFMs are found (see Fig. 2B).

An important property of EFMs is that any steady-state flux vector can be written as a linear combination of EFMs with non-negative coefficients, as shown in Eq. (3). We denote e_{rk} the relative flux of reaction r in EFM k and J_k the contribution of EFM k to the flux vector, named here EFM activity. Note here that fully reversible EFMs are divided into two independent EFMs that differ in the direction of the flux (sign of e_{rk}) and therefore its contribution (J_k) is non-negative.

$$v_r = \sum_{k=1}^{K} J_k e_{rk} J_k \geq 0, \quad r = 1,\ldots,R \tag{3}$$

The concept of Elementary Carbon Modes (ECMs) is analogous to EFMs at the carbon level. To illustrate this concept, consider the network in Fig. 2C, which represents the example metabolic network in Fig. 2A at the carbon level. The nodes of the network in Fig. 2C are now the carbon atoms of the metabolites. We assumed that metabolites in Fig. 2A involve three carbon

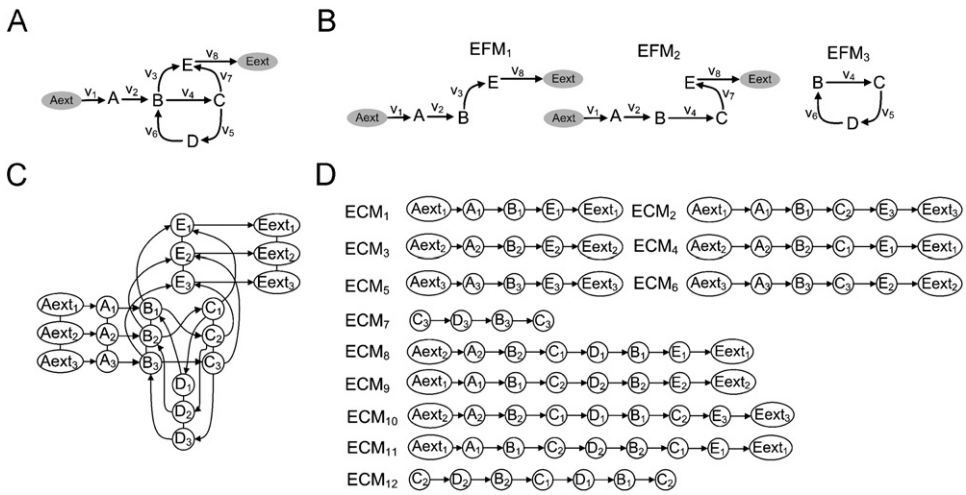

Fig. 2. (A) Example metabolic network; (B) EFMs; (C) carbon flux network and (D) ECMs.

atoms. For example, A_1, A_2 and A_3 represent the first, second and third carbon atom of metabolite A, respectively. Arcs represent underlying atom carbon movements inside a metabolic reaction. For example, in the metabolic reaction v_2, the first carbon of metabolite A (A_1) goes to the first carbon of metabolite B (B_1). It is important to observe that, while the network in Fig. 2A involves metabolite fluxes, the network in Fig. 2C represents carbon fluxes. For this reason the network in Fig. 2C is referred to as carbon flux network. Though not shown here the carbon flux network may logically involve reversible carbon fluxes, which proceed from reversible reactions in the metabolic network.

Analogously to metabolic networks, a finite set of elementary solutions can be computed for carbon flux networks, which are precisely the ECMs. Fig. 2D shows the 12 ECMs resulting from the carbon flux network in Fig. 2C. Note here that, as we have now unimolecular reactions, ECMs are simple paths or simple cycles (ECMs 7 and 12), as defined in graph theory (Bondy and Murty, 1976). Although the number of ECMs is much higher than EFMs, the fact that they constitute simple paths and cycles is an advantage from the computational point of view. However, a more detailed analysis is required to compare the computational expense of EFMs and ECMs.

As discussed in Pey et al. (2011), in analogy with EFMs and metabolic fluxes (see Eq. (3)), any distribution of carbon fluxes can be reconstructed as a linear combination of ECMs with non-negative coefficients. We denote as g_i the contribution (activity) of ECM i ($i=1,\ldots,G$) to the distribution of carbon fluxes, G being the total number of ECMs.

In the following subsections we describe how the concept of ECMs can be integrated with tracer-based metabolomics data and metabolic fluxes in a linear fashion.

2.2. ECMs and ILEs

In this sub-section we provide the mathematical formalism to link tracer-based metabolomics data of extracellular metabolites and ECMs. Extracellular metabolites are inputs/outputs metabolites defined in the metabolic network and are involved in transport fluxes. In the toy example in Fig. 2, *Aext* and *Eext* are extracellular metabolites and fluxes v_1 and v_8 their associated transport fluxes. Transport fluxes typically represent exchanges with the (physical) extracellular compartment. However, it may also represent exchanges with metabolic processes not modeled in the metabolic network under study.

Let Y_{cx} be the proportion of a labeled carbon atom x ($x=1,\ldots,X_c$) of extracellular metabolite c ($c=1,\ldots,C$; $c\in E_m$), where X_c is the number of carbons atoms of metabolite c, C is the total number of metabolites in the metabolic network and E_m is the full set of extracellular metabolites whose enrichment can be measured. Y_{cx} can be easily calculated from the experimental isotopomer distribution of metabolite c, which can be obtained via Nuclear Magnetic Resonance (NMR) (see Stephanopoulos et al., 1998).

From the full set of ECMs arriving to carbon x of extracellular metabolite c, $P_{(c,x)}$, we have a subset of them starting from a labeled carbon atom in the input, $L_{(c,x)}$. Based on the property mentioned above that ECMs reconstruct linearly the distribution of carbon fluxes, the total exchange flux through carbon x of metabolite c is equal to the sum of activities of any ECM i (g_i), such that $i\in P_{(c,x)}$. Similarly, the total transport flux through labeled carbon x of metabolite c is equal to the sum of activities of any ECM i (g_i), such that $i\in L_{(c,x)}$. Therefore, it can be observed that Y_{cx} can be reconstructed as the proportion of the total transport flux of carbon x produced by ECMs labeling carbon x, as reflected in Eq. (4). We referred to it as Enrichment constraint. Note here that the experimental measurement of Y_{cx} may be somewhat imprecise and inconsistencies may arise. For convenience, we model the error ε_{cx} as the subtraction of two non-negative variables, namely $\varepsilon_{cx}=\lambda_{cx}-\gamma_{cx}$. We introduced this error component in the numerator of Eq. (4) to hold linearity. This equation shows that ECMs and ILEs data can be linearly integrated.

$$Y_{cx}=\frac{(\sum_{i\in L(c,x)}^{G} g_i)+(\lambda_{cx}-\gamma_{cx})}{(\sum_{i\in P(c,x)}^{G} g_i)},\quad x=1,\ldots,X_c;\ \ c=1,\ldots,C;\ \ c\in E_m \tag{4}$$

To illustrate Eq. (4), Table 1 shows the Enrichment constraint when applied to the carbon flux network in Fig. 2C. It was assumed that: (i) the second carbon of the input metabolite (*Aext*) is fully labeled, while the first and third carbon atoms fully unlabeled; and (ii) 50%, 37.5% and 12.5% of the first, second and third carbon of *Eext* are labeled, respectively. Let us consider the first carbon atom of *Eext* ($Eext_1$). We have 4 ECMs producing $Eext_1$, namely $P_{(Eext,1)}=\{1,4,8,11\}$, while 2 of them come from a labeled carbon in the input, $L_{(Eext,1)}=\{4,8\}$. This was repeated for $Eext_2$ and $Eext_3$ (see Table 1).

2.3. ECMs and metabolic fluxes

The question now is how to relate the flux of ECMs (g_i) with metabolic fluxes (v_r) in a linear fashion. This relationship is referred here to as the continuity constraint.

Table 1
Constraints relating ECMs to labeling data and metabolic fluxes for the example metabolic network in Fig. 2.

ECMs-based constraints	
Carbon atom	**Enrichment constraint (Eq. (4))**
$E_{ext,}1$	$Y_{E_{ext}1}=0.5=\frac{g_4+g_8+(\lambda_{E_{ext}1}-\gamma_{E_{ext}1})}{g_1+g_4+g_8+g_{11}}$
$E_{ext,}2$	$Y_{E_{ext}2}=0.375=\frac{g_3+(\lambda_{E_{ext}2}-\gamma_{E_{ext}2})}{g_3+g_6+g_9}$
$E_{ext,}3$	$Y_{E_{ext}3}=0.125=\frac{g_{10}+(\lambda_{E_{ext}3}-\gamma_{E_{ext}3})}{g_2+g_5+g_{10}}$
Carbon flux	**Continuity constraint (Eq. (5))**
$A_1\rightarrow B_1$	$g_1+g_2+g_9+g_{11}=V_2$
$A_2\rightarrow B_2$	$g_3+g_4+g_8+g_{10}=V_2$
$A_3\rightarrow B_3$	$g_5+g_6=V_2$
$B_1\rightarrow E_1$	$g_1+g_8=V_3$
$B_2\rightarrow E_2$	$g_3+g_9=V_3$
$B_3\rightarrow E_3$	$g_5=V_3$
$B_1\rightarrow C_2$	$g_2+g_9+g_{10}+g_{11}+g_{12}=V_4$
$B_2\rightarrow C_1$	$g_4+g_8+g_{10}+g_{11}+g_{12}=V_4$
$B_3\rightarrow C_3$	$g_6+g_7=V_4$
$C_1\rightarrow D_1$	$g_8+g_{10}+g_{12}=V_5$
$C_2\rightarrow D_2$	$g_9+g_{11}+g_{12}=V_5$
$C_3\rightarrow D_3$	$g_7=V_5$
$D_1\rightarrow B_1$	$g_8+g_{10}+g_{12}=V_6$
$D_2\rightarrow B_2$	$g_9+g_{11}+g_{12}=V_6$
$D_3\rightarrow B_3$	$g_7=V_6$
$C_1\rightarrow E_1$	$g_4+g_{11}=V_7$
$C_2\rightarrow E_3$	$g_2+g_{10}=V_7$
$C_3\rightarrow E_2$	$g_6=V_7$

For each metabolic reaction r ($r=1,\ldots,R$) we define a set of pairs of metabolites $\theta_{(r)}=\{\alpha,\beta\}$ such that $s_{\alpha r}<0$, $s_{\beta r}>0$ and (α,β) exchange carbon atoms. For example, consider oxaloacetate decarboxylase (*OAADC*) reaction, which releases *CO2* and phosphoenolpyruvate (*PEP*) from oxoloacetate (*OAA*). Then, we have that $\theta_{(OAADC)}=\{(OAA, CO2);(OAA, PEP)\}$. In addition, each metabolic reaction r has a finite number of underlying carbon fluxes. Each carbon flux in a metabolic reaction r involves a different carbon atom x of metabolite α and carbon atom w of metabolite β. We denote $\theta_{(r,\alpha,\beta)}=\{x, w\}$ the set of pairs of carbon atoms x and w exchanged from metabolites α and β, respectively, in reaction r. Finally, we denote $P_{(r,\alpha,\beta,x,w)}$ the set of ECMs that involve a carbon flux in a metabolic reaction r that goes from carbon atom x in metabolite α to carbon atom w in metabolite β.

The continuity constraint establishes that the flux through a metabolic reaction (ν_r) must be equal to each of its underlying fluxes at the carbon level, which are defined by each pair of carbon atoms $(x,w)\in\theta_{(r,\alpha,\beta)}$, as reflected in Eq. (5). Importantly, each underlying carbon flux of metabolic reaction r is equal to the sum of the activity of any ECM i, such that $i\in P_{(r,\alpha,\beta,x,w)}$. Note here that a reversible carbon flux can be part of different ECMs in both forward and backward direction. For this purpose, we introduce the two-valued parameter $a_{ir\alpha\beta xw}$, which takes 1 if ECM i involves a carbon flux of metabolic reaction r (defined by a pair of carbon atoms $(x,w)\in\theta_{(r,\alpha,\beta)}$) in the positive direction, -1 otherwise.

$$\left[\left(\sum_{i\in P(r,\alpha,\beta,x,w)}^{G} a_{ir\alpha\beta xw}g_i=\nu_r,(x,w)\in\theta_{(r,\alpha,\beta)}\right),(\alpha,\beta)\in\theta_{(r)}\right],\quad r=1,\ldots,R \tag{5}$$

With this constraint it is guaranteed that the same number of molecules is produced at the carbon and metabolic level. In addition, Eq. (5) also implies that each underlying carbon flux in reaction r has the same value. This fact guarantees that we are producing each of the carbon atoms of the metabolite in the same proportion, as obviously occurs at the metabolic flux level.

Table 1 shows the Continuity constraint when applied to the example metabolic network in Fig. 2. For illustration, let us consider reaction 2. Reaction 2 involves one pair of substrate–product: $\theta_{(2)}=\{(A, B)\}$, and this pair involves three carbon fluxes, *i.e.* $\theta_{(2,A,B)}=\{(A_1, B_1); (A_2, B_2); (A_3, B_3)\}$. For continuity, the carbon flux through A_1–B_1, A_2–B_2 and A_3–B_3 must be equal to the metabolic flux, ν_2. The carbon flux through A_1 and B_1, for example, is involved in 4 out of 12 ECMs, namely $P_{(2,A,B,1,1)}=\{1,2,9,11\}$ and, therefore, its value is equal to the sum of activities of such ECMs ($g_1+g_2+g_9+g_{11}$), as observed in Table 1. Note here that as in this example all reactions are irreversible, then $a_{ir\alpha\beta xw}=1$ for all carbon fluxes and ECMs. This procedure was repeated for each of the carbon fluxes in Fig. 2C, as observed in Table 1. For the sake of simplicity, we only omitted the constraint for carbon fluxes arising from transport reactions, namely $Aext\rightarrow A$ and $E\rightarrow Ext$.

Finally, note that when Eq. (5) is applied to each reaction r ($r=1,\ldots,R$), a linear set of equations is generated. Therefore, metabolic fluxes (ν_r) and the activity of ECMs (g_i) are related in a linear fashion, as we sought.

2.4. ECMs and CBM

2.4.1. Basic constraints in CBM

Eqs. (4) and (5) generate a novel set of linear constraints that integrate tracer-based metabolomics data of extracellular metabolites and metabolic fluxes via ECMs. These constraints can be naturally coupled with the basic constraints of CBM, which are summarized below.

First, the steady-state condition is typically applied, as described in Eq. (2). In addition, it is usual to scale fluxes by measuring some transport fluxes of extracellular metabolites, as observed in Eq. (6). Note that ν_r^{exp} is the measured value of flux of reaction r and R_m is the subset of measurable metabolic fluxes.

$$\nu_r=\nu_r^{exp},\quad r=1,\ldots,R,\quad r\in R_m \tag{6}$$

It is common the introduction of an upper (u_r) and lower (l_r) bound in metabolic fluxes, as observed in Eq. (7). For irreversible reactions $l_r=0$. When no additional experimental information is available, $u_r=U$ and for reversible reactions $l_r=-U$, where U is a large positive scalar.

$$u_r\geq\nu_r\geq l_r,\quad r=1,\ldots,R \tag{7}$$

2.4.2. FVA

Interestingly, as Eqs. (2) and (4)–(7) constitute a linear set of equations, the majority of methods of CBM (Price et al., 2004), which are typically built on linear optimization, can be easily applied. In order to evaluate the effect of our novel constraints based on ECMs in reducing the underlying under-determination of metabolic networks at the flux level, we used here Flux Variability Analysis (FVA) (Mahadevan and Schilling, 2003). Note here that the integration of our set of constraints with others methods from CBM (Baughman et al., 2011; Chemler et al., 2010; Nagrath et al., 2010; Xu et al., 2011) can be easily done.

FVA determines the spectrum of numerical values for each flux in the network, while still satisfying the given constraints. For this purpose, the minimum and maximum flux for each reaction is determined:

$$\min(\max)\nu_r \tag{8}$$

As noted above, we may have an error component in the observed labeling data (Y_{cx}), which is addressed in our formulation with two non-negative variables (λ_{cx} and γ_{cx}). This error component must be accounted in our objective function when finding a solution. For this purpose, FVA is reformulated by

introducing a new term in Eq. (8) that minimizes the sum of absolute errors (1-norm or Manhattan distance). We use this distance metric error to keep linearity, as shown in Eqs. (9) and (10). This form of representing Manhattan distance is standard in Linear Programming literature (Williams, 1999). Note that U is a big scalar value, which gives primary weight to 1-norm minimization.

$$\min v_r + U \sum_{c=1, c \in Em}^{C} \sum_{x=1}^{Xc} (\lambda_{cx} + \gamma_{cx}) \tag{9}$$

$$\max v_r - U \sum_{c=1, c \in Em}^{C} \sum_{x=1}^{Xc} (\lambda_{cx} + \gamma_{cx}) \tag{10}$$

Minimizing Eq. (9) or maximizing Eq. (10) subject to Eqs. (2), (4)–(7) is a linear program, which can be solved with different software packages. We used Ilog Cplex.

2.4.3. Extension to genome-scale networks

Our set of labeling constraints presented above (Eqs. (4) and (5)) is currently limited to the computation of ECMs, which exploits in a combinatorial fashion even in medium-sized metabolic networks. This issue logically restricts the direct application of our approach to GSMNs, which are at the core of CBM.

A natural way to overcome this issue is to apply our approach to a given small metabolic sub-system and use the resulting set of constraints to restrict the solution space of GSMNs. This strategy has been similarly used to integrate kinetic models into GSMNs (Yizhak et al., 2010) and it is particularly suitable here due to the inherent linearity of both our constraints and those typically used in GSMN.

In brief, we first define a set of constraints for a given metabolic sub-system of interest that logically includes our labeling constraints. We then present a set of constraints for GSMNs and show how to link their fluxes to those in the sub-system. Finally, a suitable objective (optimization) function is re-established.

For the sake of simplicity, we use the notation and constraints presented in previous sections for the sub-system, namely Eqs. (2) and (4)–(7). Therefore, fluxes in the sub-system are denoted v_r ($r=1,\ldots,R$). We describe below variables and constraints in GSMNs.

Consider a GSMN that comprises T reactions and M metabolites. Each reaction t ($t=1,\ldots,T$) in the GSMN is associated with a flux variable w_t. Fluxes in GSMNs must satisfy steady-state condition for the set of internal metabolites (I^{gen}), as observed in Eq. (11). We denote k_{mt} the stoichiometric coefficient associated with metabolite m ($m=1,\ldots,M$) in reaction t ($t=1,\ldots, T$).

$$\sum_{t=1}^{T} k_{mt} w_t = 0, \quad m \in I^{gen} \tag{11}$$

Similarly to Eq. (6), some fluxes can be experimentally measured (Eq. (12)). In addition, fluxes are upper (u_t^{gen}) and lower bounded (l_t^{gen}). See Eq. (13).

$$w_t = w_t^{exp} \quad t = 1,\ldots,T, \quad t \in T_m \tag{12}$$

$$u_t^{gen} \geq w_t \geq l_t^{gen}, \quad t = 1,\ldots,T \tag{13}$$

Note here that some reactions in the sub-system are also included in the GSMN. We denote this set of reactions R_o. In particular, each reaction in the sub-system $r \in R_o$ has an identical reaction $t=Q_r$ in the GSMN. This is exploited to couple the equations of the sub-system (Eqs. (2), (4)–(7)) and equations of the GSMN (Eqs. (9)–(11)), namely by forcing the flux of both reactions (v_r and w_{Qr}) to be the same:

$$w_{Qr} = v_r, \quad r = 1,\ldots,R, \quad r \in R_o \tag{14}$$

Finally, the objective function (Eqs. (9) and (10)) is re-established in Eqs. (15) and (16), which applies FVA for fluxes in the GSMN

$$\min w_t + U \sum_{c=1, c \in Em}^{C} \sum_{x=1}^{Xc} (\lambda_{cx} + \gamma_{cx}) \tag{15}$$

$$\max w_t - U \sum_{c=1, c \in Em}^{C} \sum_{x=1}^{Xc} (\lambda_{cx} + \gamma_{cx}) \tag{16}$$

Again, minimizing Eq. (15) or maximizing Eq. (16) subject to Eqs. (4) and (5), (11)–(14) is a linear optimization model. With this extension of our approach, we can determine how the labeling constraints in the sub-system affect fluxes in the GSMN.

3. Results

To illustrate our approach, we first present results for a small metabolic system previously considered in Schuster et al. (2000), which involves the central routes in the degradation of glucose. We then analyze the effect of our labeling constraints in such sub-system for further restricting fluxes in the human GSMN presented in Duarte et al. (2007). For this purpose, we used simulated data. As it is not currently possible to experimentally measure the actual flux state, the predicted (based on partial experimental labeling data) and actual flux state cannot be compared.

3.1. Sub-system analysis

We applied our ECMs-based model presented above for the analysis of fluxes in the metabolic network of Fig. 3, previously published in Schuster et al. (2000). This network comprises the major part of the monosaccharide metabolism, involving glycolysis, pentose phosphate pathway (PPP) and part of gluconeogenesis, as well as an input flux of glucose (*D-Glc*) and three output fluxes of CO_2, ribose-5-phosphate (*R5P*) and pyruvate (*Pyr*). Ellipses represent metabolites and arcs reactions. We have reversible and irreversible reactions, which are distinguished in Fig. 3 by the use of a white and black dot in the middle of arcs, respectively. Cofactors (*ATP*, *ADP*, etc.) were not included in Fig. 3, since they were set as external metabolites.

The network in Fig. 3 involves 7 EFMs and 2624 ECMs. To compute EFMs and ECMs, efmtool (Terzer and Stelling, 2008) was used. Note that the number of ECMs is increased due to PPP, which introduces high complexity at the carbon level.

We assumed as input experimental data (i) the input of *D-Glc* and the output of *Pyr* at the flux level and (ii) the carbon enrichment data of entering *D-Glc*, *Pyr* and *R5P*. Therefore, Eq. (4) is applied to *D-Glc*, *Pyr* and *R5P* and Eq. (6) to the transport fluxes of *D-Glc* and *Pyr*.

In order to evaluate the effect of our ECMs-based constraints, we applied FVA for the model consisting of Eq. (2) and Eqs. (6) and (7), which are the basic constraints in CBM in (i) the presence and (ii) the absence of Eqs. (4) and (5). As noted above, we used simulated data. The procedure as to how this data was simulated is described below.

3.1.1. Data simulation

The procedure to simulate the data was the following. Firstly, we assigned a random activity to each EFM, J_k^{sim} ($k=1,\ldots,K$). Using Eq. (3), we determined metabolic fluxes, v_r^{sim}($r=1,\ldots,R$). Note here that we generated random values for the activities of EFMs, instead of random metabolic fluxes, so as to guarantee steady-state

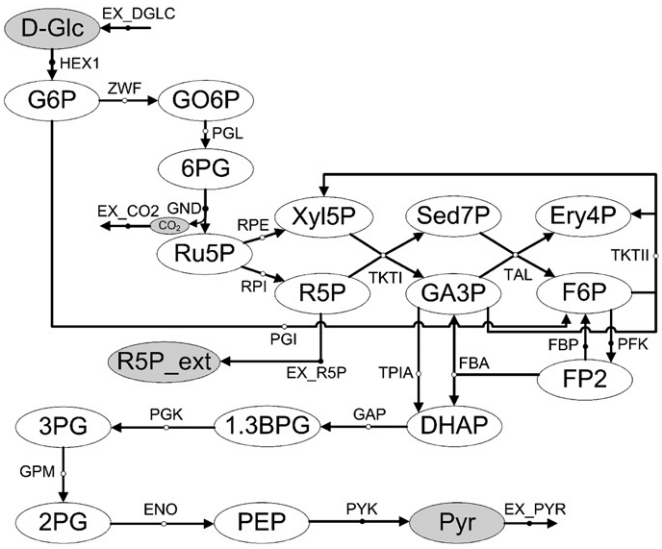

Fig. 3. Metabolic network taken from (Schuster et al., 2000). *Abbreviations of metabolites*: D-Glc, GLucose; G6P, glucose-6-phosphate; GO6P, 6-phosphoglyconate; 6PG, 6-phosphogluconate; F6P, fructose-6-phosphate; CO2, Carbon dioxide; Ru5P, ribulose-5-phosphate; Xyl5P, xylulose-5-phosphate; R5P, ribose-5-phosphate; Sed7P, sedoheptulose-7-phosphate; GA3P, glyceraldehyde-3-phosphate; Ery4P, D-erythrose-4-phosphate; FP2, fructose-1,6-bisphosphate; DHAP, dihydroxyacetone phosphate; 1.3BPG, 1,3-bisphosphoglycerate; 3PG, 3-phosphoglycerate; 2PG, 2-phosphoglycerate; PEP, phosphoenolpyruvate; Pyr, pyruvate. *Abbreviations of reactions*: EX_DCLC, transport glucose; HEX1, Hexokinase; PGI, Phosphoglucose isomerase; PFK, phosphofructokinase; FBA, fructose bisphosphate aldolase; GAP, glyceraldehyde-3-phosphate dehydrogenase; PGK, phosphoglycerate kinase; GPM, phosphoglycerate mutase; ENO, enolase; PYK, pyruvate kinase; Ex_PYR, transport pyruvate; ZWF, glucose-6-phosphate dehydrogenase; PGL, phosphogluconolactonase; GND, phosphogluconate dehydrogenase; RPE, ribulose-phosphate 3-epimerase; RPI, ribose-5-phosphate isomerase; TKTI, transketolase; TAL, transaldolase; TKTII, transketolase II; TPIA, triosephosphate isomerase; FBP, fructose-1,6-bisphosphatase; EX_CO2, transport CO2; Ex_R5P, transport ribose-5-phosphate.

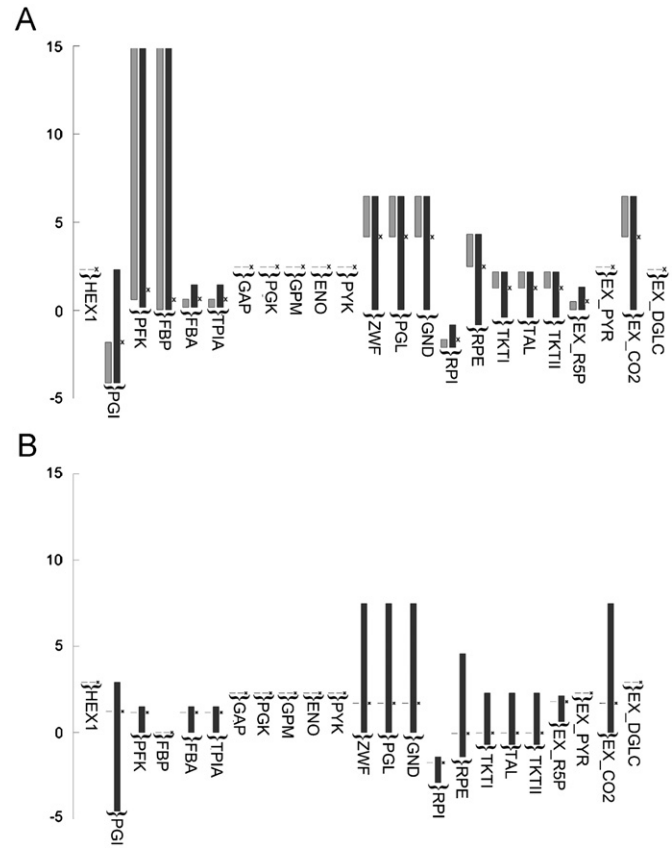

Fig. 4. Flux spectrum with and without ECMs-based constraints in the (A) presence and (B) absence of FBP reaction.

condition, Eq. (2). The knowledge of metabolic fluxes and the labeling of the input carbon source (in our case *D-Glc*) allow the calculation of the steady-state isotopomer distribution of metabolites using the method presented in Schmidt et al. (1997). See Pey et al. (2011), for a detailed summary and application of such method. As discussed above, enrichment data Y_{cx}^{sim} can be easily obtained from the isotopomer distribution of metabolites.

From the simulated data, *i.e.* v_r^{sim} and Y_{cx}^{sim}, we only introduced in our model the assumed input data, which, as noted above, is Y_{cx}^{sim} for *D-Glc*, *Pyr* and *R5P* and v_r^{sim} for transport fluxes of *D-Glc* and *Pyr*. With this information, we evaluated the effectiveness of our linear model in predicting the rest of the simulated fluxes (v_r^{sim}) of the network.

3.1.2. FVA

Fig. 4A shows the results of FVA in the presence and absence of our ECMs-based constraints (Eqs. (4) and (5)) when applied to the network in Fig. 3. For this purpose, a random instance of input data was generated with the simulation procedure described above. It was assumed that the second carbon of input glucose is fully labeled and the rest of carbons fully unlabeled. Crosses represent the actual (simulated) flux, while gray and black bars are the flux spectrum for each reaction in the presence and absence of ECMs-based constraints, respectively.

Different insights can be obtained from Fig. 4A. As noted above, the transport flux of *Pyr* and *D-Glc* is used as input data and, therefore, in both scenarios is fixed. The flux of reactions in the payoff phase of glycolysis (*GAP*, *PGK*, *GPM*, *ENO* and *PYK*) is fixed by the transport flux of *Pyr* and the transportation reaction of glucose (*HEX*1) is fixed by the transport flux of glucose in both scenarios. However, in the rest of fluxes the effect of our ECMs-based constraints is clearly observed. For example, in the second step of glycolysis (*PGI*) the flux spectrum is reduced from [2.3, −4.22] to [−2.33, −4.22], which narrows the flux interval to 30% and reveals that the flux through PGI is non-zero and negative. This emphasizes the impact of our ECMs-based constraints. Fig. 4A shows that similar results are found in many other reactions, *e.g. ZWF*, *GND*, *RPE*, etc.

The weakest prediction of our approach is found in the futile cycle formed by *FBP* and *PFK* reactions, which also constitute a cycle at the carbon level. Since this futile cycle does not involve any asymmetry at the carbon level, its flux cannot be inferred. However, in our method, this futile cycle introduces more complexity at the carbon level, which is reflected in the under-determination shown in the rest of fluxes. The nature of this under-determination requires further research. To show the role of this futile cycle in this under-determination, we removed reaction *FBP* from the metabolic network. In this new scenario, our model fully predicts the actual fluxes of the network, as observed in Fig. 4B. This clearly shows how cycles increase the complexity in metabolic networks.

The same analysis was repeated when a random Gaussian error component was directly introduced in the simulated isotopomeric data. We assumed that such Gaussian error has mean value 0 and standard deviation of 10% of (simulated) isotopomer fragment abundances (Grotkjaer et al., 2005; Meijer et al., 2009). This Gaussian error in the labeling causes a variation in the length of flux spectrums that distributes similarly to a normal random variable. Based on limited empirical evidence, we also found that the standard deviation of these variations (in the length of flux spectrums) is less than 10% with respect its length when no error component is included. This shows the robustness of our

approach to deal with errors in labeling data. A detailed discussion is found in the Supplementary Material.

3.1.3. Case study: optimal input labeling

Our linear model can be used to analyze in a systematic fashion the optimal (most informative) label configuration in the input, *i.e.* the entering input labeling that most accurately describes metabolic fluxes. This is currently a relevant topic in the field of Isotope Labeling Experiments (Crown and Antoniewicz, in press; Walther et al., 2012).

In the previous sub-section, we assumed that the second carbon of glucose was fully labeled and the rest of carbons were fully unlabeled. This is just one of 64 possible isotopomer configurations of glucose. For finding the most informative configuration, all possible 64 cases need to be analyzed. For illustration, we only evaluated 6 cases, namely each of them fully labeling one different carbon atom of glucose and the rest fully unlabeled.

To perform this analysis, we generated 100 random steady-state flux distributions, as described above. For each of the 6 possible labeling configurations, we introduced the input data into our linear model (as described above) and examined the flux spectrum in each of the 100 cases. Note that this task requires $100 \times 6 \times 23 \times 2 = 27{,}600$ linear optimization programs. This took less than an hour, since Linear Programming algorithms are very efficient.

In order to visualize the results, for each label configuration and each of the 100 simulations, we first summed the flux spectrum of all the reactions in the network, whose value is referred here to as *global flux spectrum*. Clearly, the prediction is more accurate when this value is smaller. We used boxplots for statistical analysis on the global flux spectrum of the 100 simulations for each of the six label configurations (Fig. 5A). A boxplot is a graphical representation typically used for random experiments that comprises a box and a whisker. The box extends from the first quartile to the third quartile values with an additional horizontal line that represents the median. The whiskers are vertical lines that go from each end of the box till the most extreme data value within 1.5 times the interquartile range. Points beyond the end of the whiskers are considered to be outliers and are represented by crosses.

It can be observed in Fig. 5A that the sixth carbon of glucose is the least informative label configuration, while the better prediction is found in the first three label configurations. This effect is similar if we focus on a particular reaction, *e.g. TKTII*, as noted in Fig. 5B. The same analysis was repeated when reaction *FBP* was deleted. Results are qualitatively similar, though quantitative differences are much more pronounced (see Fig. 5C and D).

3.2. Genome-scale analysis

To illustrate the extension of our ECMs-based model for GSMNs, we evaluated the effect of our labeling constraints to restrict the maximum biosynthetic capacity of non-essential amino acids in the human metabolic network presented in Duarte et al. (2007). We assumed a growth medium based on glucose as a sole carbon source, as done in metabolic studies of primary cultures. As methionine is not in the growth medium, cysteine becomes essential and therefore it has not been considered.

As introduced in the "Methods" section, our labeling constraints (Eqs. (4) and (5)) are applied to a small sub-system and then fluxes in the sub-system are linked to those in GSMNs (Eq. (14)). For the sake of simplicity, we used the sub-system presented in Fig. 3. This sub-system includes the biosynthesis of *R5P* and *Pyr*, which are key precursors of nucleotides and some amino acids. Note here that for the production of serine (*L-Ser*) and glycine (*Gly*), 3-Phosphoglycerate (3*PG*) is required as precursor. For completeness, we include a further transport reaction of 3*PG* in Fig. 3.

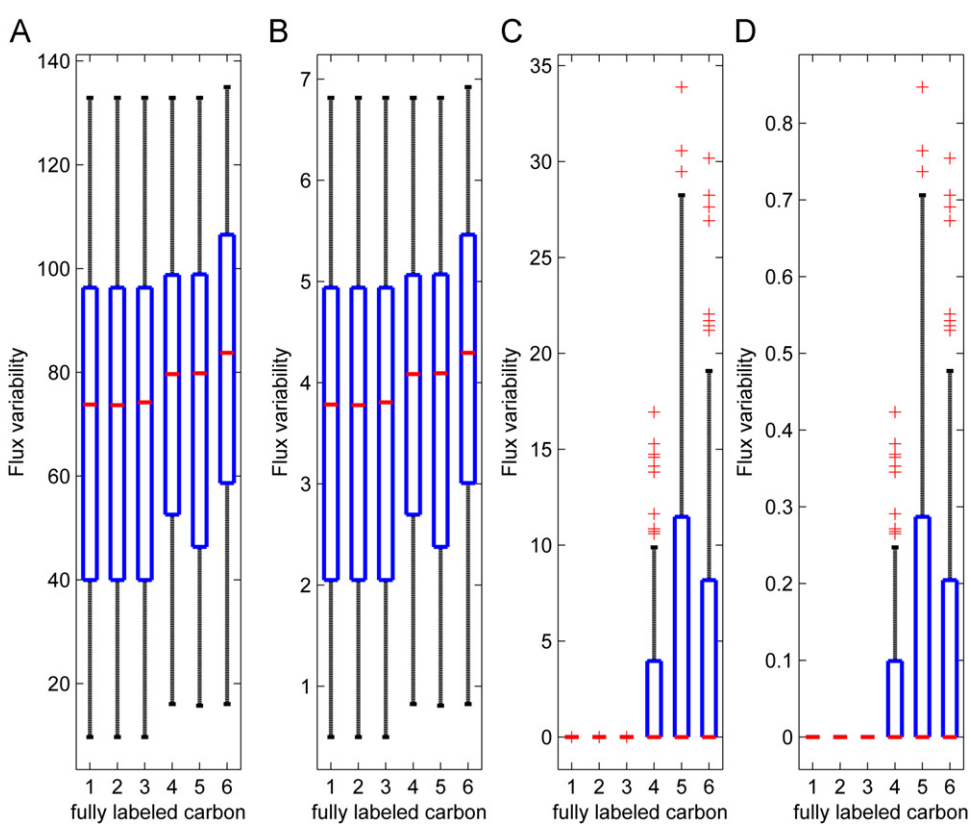

Fig. 5. Boxplot analysis of 100 random simulations of (A) global flux spectrum for each label configuration; (B) the flux spectrum for TKTII reaction; (C) same as (A) with reaction FBP deleted; (D) same as (B) with reaction FBP deleted.

We assumed as input experimental data (i) the input of D-*Glc* and (ii) the carbon enrichment data of entering D-*Glc*, *Pyr*, *R5P* and 3*PG*. Therefore, Eq. (4) is applied to D-*Glc*, *Pyr*, *R5P* and 3*PG* and Eq. (12) to the transport fluxes of D-*Glc*. We used simulated data as described below. Again, in order to see the relevance of our labeling constraints, the effect of noise is neglected in this section.

3.2.1. Data simulation and performance measure

As in Section 3.1.1, we first assigned a random activity to each EFM in the sub-system, J_k^{sim} ($k=1,\ldots,K$). Using Eq. (3), we determined metabolic fluxes, $v_r^{sim}(r=1,\ldots,R)$. As noted above, with the knowledge of metabolic fluxes and the labeling of the input carbon source we determined the steady-state isotopomer distribution of metabolites using the method presented in Schmidt et al. (1997). Enrichment data Y_{cx}^{sim} can be easily obtained from the isotopomer distribution of metabolites. From the simulated data, v_r^{sim} and Y_{cx}^{sim}, we only introduced in our model the assumed input data, which, as noted above, is Y_{cx}^{sim} for D-*Glc*, *Pyr*, *R5P* and 3*PG* and v_r^{sim} for the transport flux of D-*Glc*.

Assuming that $w_{Qr}=v_r^{sim}$, $r=1,\ldots,R$, $r\in R_o$, *i.e.* reactions in the GSMN also included in the sub-system (R_o) are forced to be equal to its simulated value (v_r^{sim}), we can determine via linear programming the maximum theoretical flux for any reaction t ($t=1,\ldots,T$) in the GSMN. This is denoted as W_t^{max}. The same can be done if we only introduce partial information: (i) assumed known extracellular fluxes (in our case D-*Glc*); and (ii) assumed known extracellular fluxes and labeling data (in our case D-*Glc*, *Pyr*, *R5P* and 3*PG*). We denote them W_t^S and W_t^L, respectively. The comparison of these two parameters (W_t^S and W_t^L) in relation with the maximum theoretical flux (W_t^{max}) will give us the effect of our labeling constraints in any reaction t. For this purpose, we used the parameter in Eq. (17) (ϕ_t), similarly to Suthers et al. (2007). Note here that as $W_t^S \geq W_t^L \geq W_t^{max}$, the effect of our labeling constraints will be particular pronounced as ϕ_t near 0, and ineffective when near 100.

$$\phi_t = \frac{w_t^L - w_t^{max}}{w_t^S - w_t^{max}} \times 100 \tag{17}$$

3.2.2. Performance analysis

We evaluated the effect of our novel constraints to restrict the maximum biosynthetic flux for non-essential amino acids via ϕ_t. We applied the procedure described above to generate 100 random scenarios, assuming that the second carbon of input glucose is fully labeled and the rest of carbons fully unlabeled, as in previous section. Results are shown in Fig. 6A using boxplots. It can be observed that in 6 out of 9 non-essential amino acids labeling constraints do have effect, ranging from an average ϕ_t of approximately 25–75%. This shows the predictive power of our ECMs-based approach even in GSMNs. Note here that the effect of our labeling constraints is irrelevant in L-Aspartate (L-*Asp*), L-Arginine (L-*Arg*) and L-Asparagine (L-*Asn*). The use of a different labeling configuration in the input glucose provides further insights. In particular, when the fourth carbon of input glucose is fully labeled and the rest of carbons fully unlabeled, the maximum biosynthetic capacity of L-*Arg* is further restricted, as observed in Fig. 6B. This is consonance with previous works that emphasize that the most informative labeling configuration may change depending on the target flux to be inferred (Crown and Antoniewicz, in press; Walther et al., 2012).

4. Discussion and conclusions

In this article we have presented a novel set of constraints based on Elementary Carbon Modes that integrate tracer-based metabolomics data of extracellular metabolites and metabolic fluxes. To illustrate the relevance of these constraints, a Flux Variability Analysis (FVA) approach was applied to the metabolic network presented in Schuster et al. (2000). The addition of our ECMs-based constraints substantially reduced the under-determination resulting from a standard application of FVA.

Our ECMs-based approach is conceptually different to any other existing mathematical approach in Isotope Labeling Experiments, the most important property being its underlying linearity. Linearity is relevant from different perspectives. Importantly, linearity facilitates integration with methods of CBM (Price et al., 2004) which typically build on linear optimization. The success of linear optimization techniques in CBM is the existence of powerful algorithms and software packages for solving them, which allows the analysis of different theoretical questions that require high computational demand. An example of this is the analysis of the optimum labeling of entering glucose accomplished in the "Results" section, which is currently a relevant topic in Isotope Labeling Experiments. In summary, the use of a linear model present advantages at the theoretical, modeling and computational level.

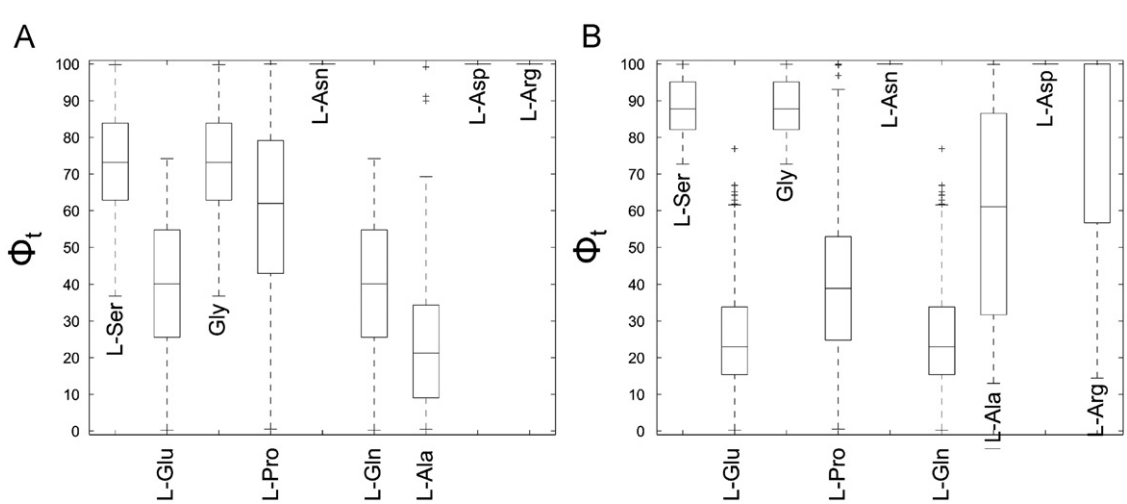

Fig. 6. Boxplot analysis of 100 random simulations of ϕ_t for non-essential amino acids when (A) the second carbon of input glucose is fully labeled and the rest of carbons fully unlabeled and (B) the fourth carbon of input glucose is fully labeled and the rest of carbons fully unlabeled. L-Ser, L-Serine; L-Glu, L-Glutamate; Gly, Glycine; L-Pro, L-Proline; L-Asn, L-Asparagine; L-Gln, L-Glutamine; L-Ala, L-Alanine; L-Asp, L-Aspartate; L-Arg, L-Argnine.

Our ECMs-based approach is currently limited to the computation of ECMs, which suffers from a combinatorial explosion as the network size increases. We overcame this issue by applying our approach to a specific tractable metabolic sub-system and using the resulting set of equations to restrict the solution space of GSMNs. This strategy has been previously used to integrate kinetic models into GSMNs (Yizhak et al., 2010). We evaluated the relevance of our extended model to restrict the maximum biosynthetic capacity of non-essential amino acids in the human GSMN presented in Duarte et al. (2007). Again, we showed the capability of our approach to reduce the inherent under-determination metabolic systems even in GSMNs.

Our ECMs-based approach currently faces several improvements. Firstly, our linear set of labeling constraints requires the knowledge of the enrichment of individual carbon atoms in (measured) extracellular metabolites. As noted above, this can be easily calculated from the experimental isotopomer distributions via Nuclear Magnetic Resonance. This type of positional labeling information generally cannot be obtained with Mass Spectrometry. Mass Spectrometry gives the total isotopic enrichment of a metabolite or metabolite fragment, which generally contains multiple carbons. Therefore, our approach must be reformulated to integrate labeling data from Mass Spectrometry.

Secondly, our enrichment constraints (Eq. (4)) are applied to metabolites defined as extracellular, namely input and output metabolites involved in a transport reaction in the metabolic network. Therefore, their application to internal metabolites is not currently possible. This is due to the fact that linear ECMs do not capture the full set of pathways producing the carbon atoms of internal metabolites. As tracer-based metabolomics data is also available for metabolites typically considered as internal metabolites, *e.g.* some intermediates of glycolysis and TCA cycle (Yang et al., 2006), further research is required to generalize our enrichment constraint to internal metabolites.

In summary, we think our ECMs-based approach constitutes an important step forward for the integration of "omics" data into metabolic networks, a hot topic in Systems Biology.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.ymben.2012.03.011.

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